R16

Code No: 136CX

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, May - 2019
MOBILE COMPUTING
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a)	List out the different types of logical channels that are used in GSM.	[2]
b)	What are the characteristics of communication devices?	[3]
c)	What are the drawbacks of FDMA?	[2]
d)	What is tunneling?	[3]
e)	What is meant by timeout freezing in mobile transport layer?	[2]
f)	What are the advantages of hoarding data at the mobile device?	[3]
g)	What is data synchronization?	[2]
h)	Explain the functions of Pull based mechanisms in data delivery mechanisms.	[3]
i)	What are the features of WAP?	[2]
j)	What are the properties of MANET?	[3]

PART - B

(50 Marks)

2. Explain the system architecture of GSM with neat sketch. [10]

OR

3. Explain the protocol architecture of DECT. [10]

4.a) Describe about the MAC layer of IEEE 802.11.
b) Explain about the CDMA and about its versions. [5+5]

OR

5.a) What is the reason for the failure of MAC schemes in wireless networks. What is done in wired networks to avoid this effect?
b) Explain about the Dynamic Host Configuration Protocol. [5+5]

6.a) What are the implications of traditional TCP on mobility? Discuss any one strategy that attempts to increase TCP performance in wireless and mobile environment.
b) Explain about the data recovery process. [5+5]

OR

7.a) Explain the issues related to quality of service.
b) Explain the header format of Mobile TCP. [5+5]

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8.a) Explain about the broadcast models.
b) Explain about the Directory method in selective tuning technique. [5+5]

OR

9.a) Explain about the data synchronization protocol.
b) Explain about the classification of data delivery mechanisms. [5+5]

10.a) Explain about the Wireless Session protocol of the session layer.
b) Explain about the MAC layer of Bluetooth. [5+5]

OR

11.a) Describe the architecture of Windows CE.
b) Explain the classification of routing algorithms. [5+5]

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